DSA Experiment 7

Title: Evaluation of postfix expression

Name- Mokshada Hejib **PRN**-B25ET1013 **Div**-SY1 A

```
#include<stdio.h>
#include<ctype.h>
#define  MAX  5
int stack[MAX];
int top=-1;
void push(int x)
{
stack [++top]=x;
}
int pop()
{
   return stack[top--];
}
int evaluatepostfix(char*exp)
{
   int i;
   for(i=0;exp[i]!='\0';i++)
   {
      char ch=exp[i];
      if(isdigit(ch))
      {
         push(ch-'0');
      }
      else
      {
         int val1=pop();
         int val2=pop();
         switch(ch)
         {
         case'+':
         push(val1+val2);
         break;
         case'-':
         push(val1-val2);
         break;
```

```
            case'*':
            push(val1*val2);
            break;
            case'/':
            push(val1/val2);
            break;
            }
        }
    }
return pop();
}
int main()
{
    char exp[100];
    printf("Enter postfix expression:");
    scanf("%s",exp);
    printf("Result=%d\n",evaluatepostfix(exp));
    return 0;
}
```

OUTPUT:

Output

```
Enter postfix expression:69*
Result=54
```

Output

```
Enter postfix expression:53+
Result=8
```